CS212 Review Problems: Graphics and Simulation (Lectures 19–22)

Multiple Choice Questions

1. What is the purpose of `paintComponent(Graphics g)`?
   - A. To handle button clicks
   - B. To draw the component
   - C. To create a window
   - D. To store data
2. Why do we call `super.paintComponent(g)`?
   - A. To speed up drawing
   - B. To clear the panel before drawing
   - C. To add components
   - D. To resize the window
3. What does `repaint()` do?
   - A. Immediately draws shapes
   - B. Calls paintComponent directly
   - C. Requests that the component be redrawn
   - D. Clears the model
4. Why should we not change state inside `paintComponent`?
   - A. It is too slow
   - B. It breaks separation of concerns
   - C. It prevents repaint from working
   - D. It causes compilation errors
5. What is the role of the model in a graphics program?
   - A. Draw shapes
   - B. Handle events
   - C. Store and update state
   - D. Create windows

6. Why is it better for the GUI to draw from the model rather than store its own data?

   A. It makes drawing faster
   B. It avoids needing repaint
   C. It keeps responsibilities clearly separated
   D. It reduces memory usage

7. What is the main purpose of a `Timer` in a GUI program?

   A. To create parallel threads
   B. To repeatedly trigger updates over time
   C. To delay user input
   D. To store elapsed time

8. Why is a `while(true)` loop not appropriate in a GUI?

   A. It runs too fast
   B. It uses too much memory
   C. It blocks the GUI from responding
   D. It prevents repaint

9. What is the key idea behind animation?

   A. Drawing once
   B. Updating state and redrawing repeatedly
   C. Using multiple threads
   D. Storing images

10. What role does iteration play in graphics programs?

    A. It handles events
    B. It creates windows
    C. It draws multiple objects
    D. It resizes components

11. Why do we separate model and GUI?

    A. To reduce code size
    B. To simplify layout
    C. To improve design and maintainability
    D. To speed up execution

12. What determines what appears on the screen?

    A. The order of method calls
    B. The contents of the model
    C. The size of the window
    D. The layout manager

Short Answer

a. What is the relationship between the model and what is drawn on the screen?

b. What two types of events exist in our programs?

c. Why does animation require repeated calls to `repaint()`?

Programming Problem 1 (Simple Drawing)

Write a program with a custom `JPanel` that draws:

- Three circles in different positions
- Each circle a different color

Programming Problem 2 (Structured Drawing)

Write a program where:

- The model stores a list of rectangles (positions only)
- The GUI draws all rectangles
- A button adds a new rectangle at a random position

Programming Problem 3

Design a simulation with the following behavior:

Features:

- Objects appear at random positions
- Each object slowly drifts downward
- Objects disappear when they reach the bottom

Tasks:

1. Define a class representing one object (position only)
2. Define a model class that:
   - stores a list of objects
   - updates all objects over time
   - removes objects when needed
3. Write GUI code that:
   - uses a `Timer`
   - updates the model
   - redraws the panel